Stellaverb

By Adrian Lam

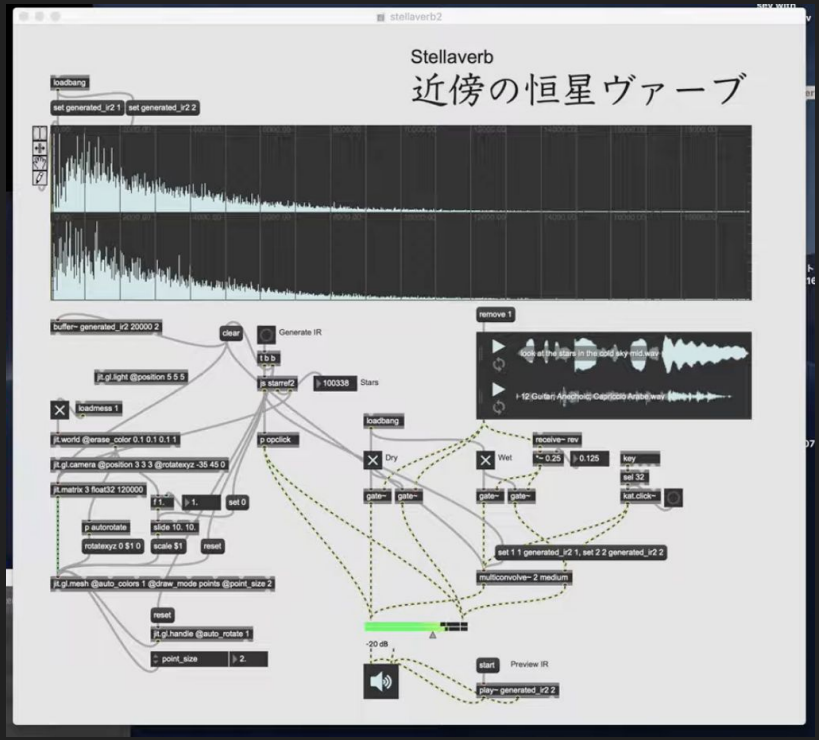
What is Stellaverb?

- Convolution plugin
- Generates IR based on ESA Gaia star catalogue
- Idea is to simulate the acoustic effect of sound propagating to the stars and back



Inspiration

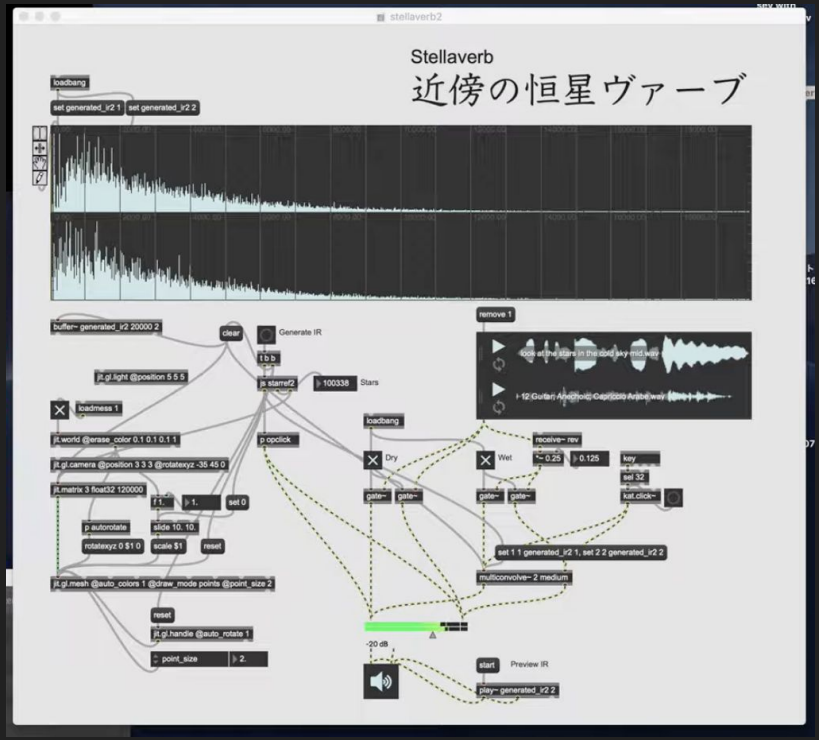
- Stellaverb Max MSP patch by Japanese electronic musician and sound designer Katsuhiro Chiba
- Implemented using the Hipparcos Star Catalogue (110,000 stars)
- For each star, map its distance to delay time and amplitude attenuation, its direction to parameters of 3D panning



Inspiration

My project expands on Stellaverb by

- implementing it as a VST plugin
- using the larger, more precise Gaia star catalogue instead (2000 million stars)
- adding adjustable parameters for IR generation
- adding additional signal processing effects such as filters and saturation



Creating Pre-processed Gaia Subset

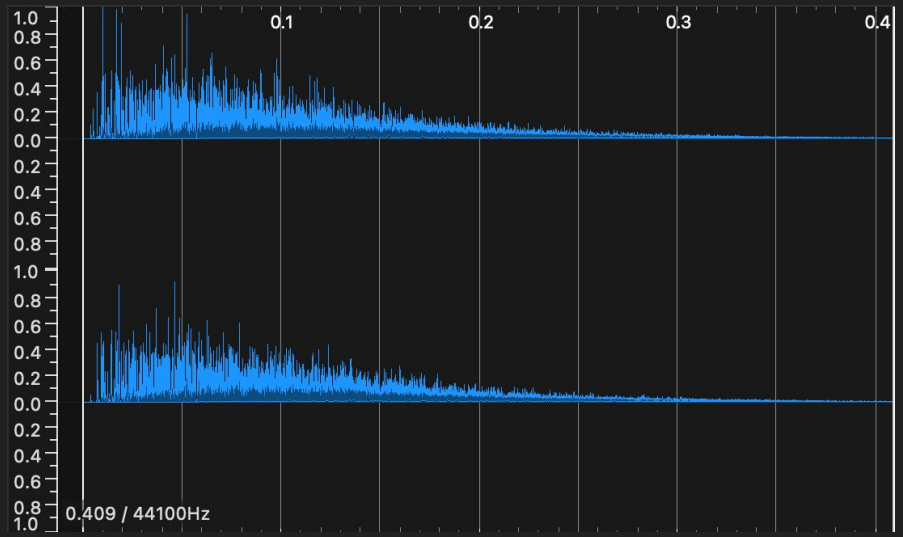
- Query and save a subset of star data using astropy (stars within about 143 parsecs with reliable measurements, ~900,000 stars)
- Used Python to generate IRs, testing different mappings
- Pre-processed star data into normalized amplitude, left and right channel gain, and distance values and packed into a binary file



Creating Pre-processed Gaia Subset

Final mappings of each star to create an impulse response

- Distance -> delay time
- Direction -> stereo panning
- Brightness -> amplitude



IR Generation Parameters

Users can customize the following aspects of IR generation

- IR Length
- Decay Time
- Min and Max Distance (of the stars used in IR generation)
- Min and Max Brightness (of the stars used in IR generation)

Main IR Generation Loop

```
for (const auto& s : stars)
{
    if (s.distance < minDistance or s.amplitude < minBrightness or s.amplitude > maxBrightness) {
        continue;
    }
    if (s.distance > maxDistance)
    {
        break; // Since the dataset is sorted by distance, we don't have to check any of the stars after
    }

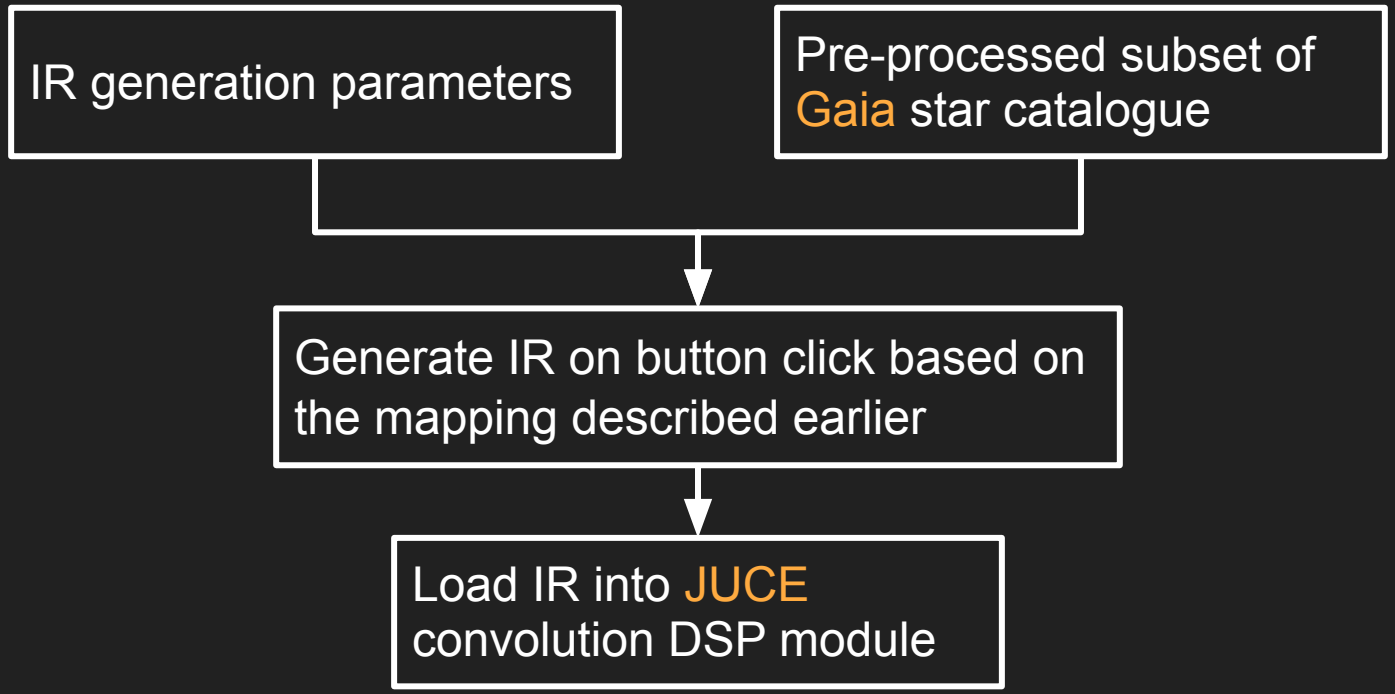
    const float norm = (s.distance - minDistance) / (maxDistance - minDistance);
    int idx = (int)(norm * (irLengthSamples - 1));

    if (idx >= 0 && idx < irLengthSamples)
    {
        float sample = s.amplitude;

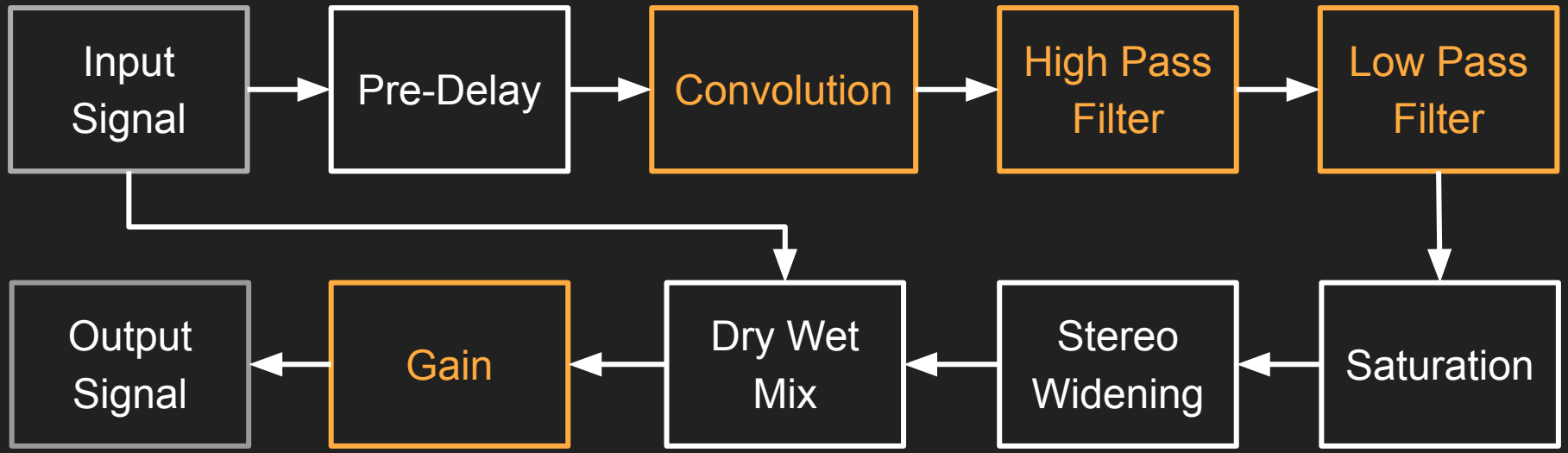
        left[idx]  += sample * s.left;
        right[idx] += sample * s.right;
    }

}
```

Flow Diagram - IR Generation



Flow Diagram - Signal Flow



Orange boxes mean they use the corresponding JUCE DSP modules

Live Demo